Efficacy and safety of weekly paclitaxel with or without ramucirumab as second-line therapy for the treatment of advanced gastric or gastroesophageal junction adenocarcinoma (RAINBOW-Asia): a randomised, multicentre, double-blind, phase 3 trial

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Summary

Background In the global phase 3 RAINBOW study, ramucirumab plus paclitaxel significantly improved overall survival compared with placebo plus paclitaxel in patients with advanced gastric or gastro-oesophageal junction (GEJ) adenocarcinoma. RAINBOW-Asia, a bridging study with similar design to RAINBOW, aimed to evaluate the efficacy and safety of ramucirumab plus paclitaxel for advanced gastric or GEJ adenocarcinoma in Asian, predominantly Chinese, patients.

Methods RAINBOW-Asia was a randomised, double-blind, placebo-controlled, phase 3 trial done at 32 centres in China, Malaysia, the Philippines, and Thailand. Adult patients (≥18 years) with metastatic or locally advanced, unresectable gastric or GEJ adenocarcinoma who previously received fluoropyrimidine–platinum-based chemotherapy were randomly assigned with a centralised interactive web response system in a 2:1 ratio to receive ramucirumab 8 mg/kg or placebo intravenously on days 1 and 15 plus paclitaxel 80 mg/m^2 intravenously on days 1, 8, and 15 of every 28-day cycle. Randomisation was stratified by Eastern Cooperative Oncology Group performance status and presence of peritoneal metastases. The co-primary endpoints were progression-free survival and overall survival. Efficacy analyses were done in the intention-to-treat population, and safety analysis included patients who received at least one dose of study treatment. This trial is registered with ClinicalTrials.gov, NCT02898077, and has been completed.

Findings Between March 2, 2017, and June 30, 2020, 440 patients were randomly assigned to receive ramucirumab plus paclitaxel (n=294) or placebo plus paclitaxel (n=146). Median progression-free survival was 4·14 months (95% CI 3·71–4·30) in the ramucirumab plus paclitaxel group compared with 3·15 months (2·83–4·14) in the placebo plus paclitaxel group (hazard ratio [HR] 0·765, 95% CI 0·613–0·955, p=0·0184). Median overall survival was 8·71 months (95% CI 7·98–9·49) in the ramucirumab plus paclitaxel group and 7·92 months (6·31–9·10) in the placebo plus paclitaxel group (HR 0·963, 95% CI 0·771–1·203, p=0·7426). The most common grade 3 or worse treatment-emergent adverse events were decreased neutrophil count (159 [54%] of 293 patients in the ramucirumab plus paclitaxel group *vs* 56 [39%] of 145 in the placebo plus paclitaxel group), decreased white blood cell count (127 [43%] *vs* 42 [29%]), anaemia (46 [16%] *vs* 24 [17%]), hypertension (21 [7%] *vs* nine [6%]), and febrile neutropenia (18 [6%] *vs* one [<1%]).

Interpretation These findings, along with the results from RAINBOW, support the use of ramucirumab plus paclitaxel as second-line therapy in a predominantly Chinese population with advanced gastric or GEJ adenocarcinoma.

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For the Chinese translation of the abstract see Online for appendix 1

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Introduction

Gastric cancer is a common malignancy worldwide.[1] The incidence of gastric cancer varies widely, with approximately 60% of gastric cancer cases occurring in east Asia.[1] East Asia also has the highest estimated mortality rates from gastric cancer.[1] The disparity in incidence and mortality between western and eastern gastric cancer populations might be associated with differences in the risk factors, clinical presentations, and treatment strategies.[2]

China accounts for 43·9% of the global incidence of gastric cancer and 48·6% of all gastric cancer-related deaths.[1,3] Gastric cancer is the second most prevalent cancer in China and the third leading cause of cancer-related deaths among men and women, with 478 508 new cases and 373 789 deaths reported in 2020.[3] At the time of diagnosis, more than 80% of patients in China are at an advanced stage of the disease,[2] and 5-year survival among patients with advanced or metastatic gastric cancer has

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Research in context

Evidence before this study

Before the study began, we searched PubMed and presentations at major oncology congresses for articles or abstracts published in English between Oct 12, 2011, and Dec 17, 2015, that reported primary results of clinical trials assessing second-line systemic treatment in patients with advanced gastric cancer. Publications describing multimodality treatment with surgery or radiotherapy were excluded. We used the search terms "stomach neoplasms" AND ("advanced" OR "recurrent" OR "metastatic") AND ("drug therapy" OR "chemotherapy") AND ("phase III" OR "phase 3" OR "randomised"), and we filtered the search by "clinical trial". Several new targeted therapeutic agents have been tested as second-line treatments for advanced gastric cancer, but these have mostly failed to produce the desired efficacy and safety results in large clinical trials. A notable exception is ramucirumab, as a single agent or in combination with paclitaxel, which is the only approved second-line regimen for advanced gastric cancer in the USA and European Union, based on findings of the REGARD and RAINBOW studies. The REGARD and RAINBOW studies did not enrol patients from mainland China, and various studies have shown that geographic and ethnic differences between Asian and non-Asian populations, and also within Asian populations, potentially affect the presentation and outcome of gastric cancer. Therefore, this bridging study (RAINBOW-Asia) was designed to evaluate the efficacy and safety of ramucirumab in combination with paclitaxel in Asian, predominantly Chinese, patients with advanced gastric or gastro-oesophageal (GEJ) adenocarcinoma who had progressed on standard therapy with platinum and fluoropyrimidines.

Added value of this study

To date, no second-line therapy has been approved for advanced gastric or GEJ adenocarcinoma in China; the only options are taxane or irinotecan monotherapy or paclitaxel/fluoropyrimidine-based chemotherapy regimens, which have poor outcomes. The development of novel therapeutic options for Chinese patients with advanced gastric cancer is therefore urgently needed. The efficacy data and safety profile of ramucirumab in combination with paclitaxel in the RAINBOW-Asia study, which included a predominantly Chinese population, demonstrate a favourable risk–benefit profile, consistent with the global RAINBOW study. Therefore, this combination might be an effective treatment option for Chinese patients with advanced gastric or GEJ adenocarcinoma in the second-line setting.

Implications of all the available evidence

RAINBOW-Asia provides evidence that ramucirumab plus paclitaxel might be a treatment option for Chinese patients with advanced gastric or GEJ adenocarcinoma in the second-line setting.

been reported to be 5·5%.[4] Platinum–fluoropyrimidine-based chemotherapy regimens are the standard treatment for advanced gastric cancer in the first-line setting.[5] Nivolumab combined with FOLFOX (leucovorin, fluorouracil, and oxaliplatin) or XELOX (capecitabine plus oxaliplatin) is recommended for patients with a PD-L1 combined positive score of 5 or more.[2] However, no standard regimen has been established for gastric cancer in the second-line setting in China. Monotherapy regimens—eg, docetaxel, paclitaxel, or irinotecan—are recommended as grade 1 preferred options.[2] For patients with metastatic microsatellite instability-high or deficient mismatch repair solid tumours, pembrolizumab is also included as an option for second-line therapy.[2] The reported outcomes for current treatment options for second-line treatment of gastric cancer in China are not promising.[6–8] Thus, there is an urgent unmet need for new treatment options for Chinese patients with advanced gastric cancer.

Ramucirumab is a humanised monoclonal antibody that specifically targets the extracellular domain of the VEGF receptor 2, inhibiting the binding of VEGF-A, VEGF-C, and VEGF-D ligands.[9,10] The REGARD study, an international, randomised, placebo-controlled, double-blind, phase 3 trial, showed the benefit of ramucirumab in patients with advanced gastric or gastro-oesophageal junction (GEJ) adenocarcinoma progressing after first-line chemotherapy.[11] The RAINBOW study was a global, randomised, placebo-controlled, double-blind, phase 3 trial conducted in 27 countries.[8] Patients with advanced gastric or GEJ adenocarcinoma treated with ramucirumab plus paclitaxel showed significant overall survival benefit over those treated with placebo plus paclitaxel (median overall survival 9·6 months *vs* 7·4 months, respectively; hazard ratio [HR] 0·807 [95% CI 0·678–0·962]; p=0·017). A significant progression-free survival benefit was also shown for ramucirumab plus paclitaxel over placebo plus paclitaxel (median 4·4 months *vs* 2·9 months; HR 0·635 [95% CI 0·536–0·752]; p<0·0001).

RAINBOW-Asia was designed as a bridging study of RAINBOW to evaluate the efficacy and safety of ramucirumab in Asian, predominantly Chinese, patients with advanced gastric or GEJ adenocarcinoma who have progressed on standard therapy with platinum and fluoropyrimidines, and to determine consistency with the benefit observed in the global population of RAINBOW.

Methods

Study design and participants

RAINBOW-Asia was a multinational, randomised, double-blind, placebo-controlled, phase 3 study conducted at

32 centres in China, Malaysia, the Philippines, and Thailand (appendix 2 pp 9–10). RAINBOW-Asia was designed as a bridging study to corroborate that the clinical data for efficacy and safety with ramucirumab and paclitaxel in Asia are similar to those in studies conducted elsewhere; there is no consensus on the statistical criteria for bridging studies, with successful criteria for such studies being set through consultation with the local regulatory authority.[12]

Eligibility criteria included age of 18 years or older; having histopathologically or cytologically confirmed advanced gastric or GEJ adenocarcinoma; metastatic disease or locally advanced, unresectable disease; documented objective radiographic or symptomatic disease progression during or within 4 months of the last dose of first-line platinum–fluoropyrimidine-based chemotherapy; Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1; at least one measurable lesion based on Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1; resolution of all clinically significant toxic effects of chemotherapy, surgery, radiotherapy, or hormonal therapy (with the exception of alopecia), except where otherwise noted in this eligibility criteria, to grade 1 or less according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03; adequate organ function; and urinary protein of less than or equal to 1+ on dipstick or routine urinalysis. Exclusion criteria included having squamous cell or undifferentiated gastric cancer; any significant gastrointestinal bleeding or any significant venous thromboembolism within 3 months before randomisation; or perforation, fistulae, or any arterial thromboembolic event within 6 months before randomisation; or uncontrolled arterial hypertension; or any first-line chemotherapy other than platinum and fluoropyrimidine for advanced gastric or GEJ adenocarcinoma; or any previous systemic therapy targeting VEGF or the VEGF receptor signalling pathways.

The study was approved by the institutional review board of each participating centre and was done in accordance with the principles of the Declaration of Helsinki and the Good Clinical Practice Guidelines of the International Conference on Harmonisation. All patients provided written informed consent.

Randomisation and masking

After confirming eligibility, patients were registered via an interactive web response system. Each patient was assigned a unique study identification number and randomly assigned in a 2:1 ratio to ramucirumab plus paclitaxel or placebo plus paclitaxel. A stratified permuted block randomisation method was used to balance the treatment allocation. Randomisation was stratified by ECOG performance status (0 *vs* 1) and peritoneal metastases (yes *vs* no). Patients, study personnel at investigational sites, study investigators, and the study sponsor were all blinded to treatment.

Procedures

Patients received either ramucirumab 8 mg/kg or placebo intravenously on days 1 and 15 plus paclitaxel 80 mg/m² intravenously on days 1, 8, and 15 of every 28-day cycle. Treatment was continued until disease progression, unacceptable toxicity, protocol non-compliance, or consent withdrawal. The criteria for dose modification or discontinuation of ramucirumab and paclitaxel are summarised in appendix 2 (p 2).

See Online for appendix 2

Tumour assessment was done using CT scans or MRI according to RECIST version 1.1 every 6 weeks (±5 days) following the first study dose for 6 months, and every 9 weeks (±5 days) thereafter until disease progression. During post-discontinuation follow-up, all patients were followed for survival (every 8 weeks [+0 to 7 days]) for as long as the patient was alive, or until study completion.

Safety assessments included adverse events, laboratory abnormalities (haematology, coagulation, chemistry, and urinalysis), and vital signs. Safety data were monitored throughout the course of the study. Adverse events were coded using the Medical Dictionary for Regulatory Activities; clinical laboratory toxicity and adverse events were graded using the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03.

Patient-reported outcomes were assessed at baseline, within 3 days before the start of each cycle, and at the short-term 30-day safety follow-up visit, using the European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire Core 30 (EORTC QLQ-C30, version 3.0) and the EuroQol 5-Dimension-3 Level Questionnaire (EQ-5D-3L).[13,14]

Outcomes

The co-primary endpoints were progression-free survival and overall survival. Progression-free survival was defined as the time from randomisation to first documented disease progression or death, whichever occurred first. Overall survival was defined as the time from randomisation to death from any cause. The study was designed with overall survival as the sole primary endpoint in earlier versions (original protocol, amendment a and b) of the protocol, and progression-free survival was added as a co-primary endpoint later in protocol amendment (c) with the purpose of accelerating the New Drug Application submission due to the urgent need for second-line gastric cancer treatment in China, in alignment with the Chinese regulatory authority. An interim analysis was also added in protocol amendment (c) to assess the primary endpoints with a predefined success criterion, namely, a stratified HR for overall survival was less than 1 and a nominal two-sided p-value for progression-free survival was less than 0·05.

Secondary endpoints were time to progression, defined as the time from randomisation to first documented disease progression; objective response rate, defined as the proportion of patients with a best overall response of complete response or partial response; and duration of

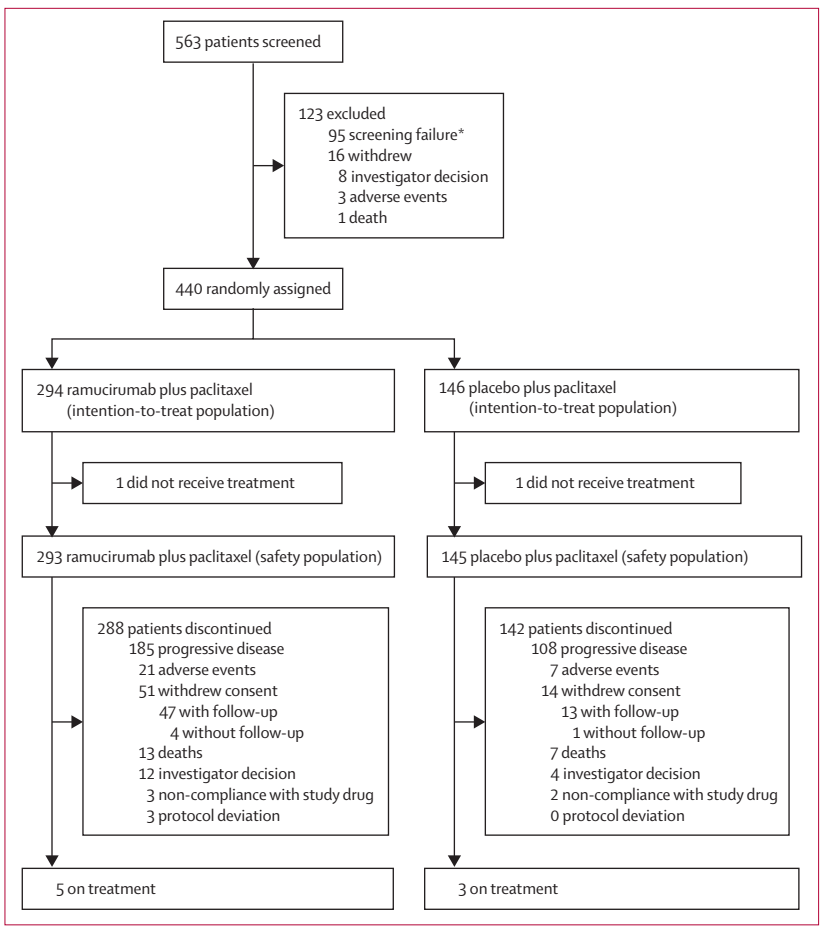

Figure 1: **Trial profile**
*Patients did not meet inclusion or exclusion criteria.

objective response, defined as the time from complete response or partial response (whichever was first recorded) until disease progression or death; safety; and patient-reported outcomes.

Statistical analysis

The planned sample size of 450 patients was based on the following assumption: 336 overall survival events were required to provide at least 80% probability of showing consistency with the results of the RAINBOW trial. More specifically, under the assumptions of a 2:1 randomisation ratio, exponentially distributed overall survival, an underlying overall survival HR of 0·81, a median overall survival of 10·5 months in the placebo group and a 25% censoring rate, and an effect size δ=1/HR–1, a sample size of 450 patients (ie, 336 deaths) would assure that $P(\delta_{RAINBOW-Asia}>0{\cdot}5*\delta_{RAINBOW})\geq 80\%$, where $\delta_{RAINBOW}=0{\cdot}235$ was estimated from the observed HR of 0·81 in RAINBOW[8] and $\delta_{RAINBOW-Asia}$ was the observed HR in the current study. No type I or type II error was specified, and all analyses were intended as descriptive; in particular, p-values were nominal and not used for inferential purpose.

An interim analysis was planned after at least 256 progression-free survival events had occurred. The number of events was determined such that the probability of obtaining a nominal log-rank p-value of less than 0·025 for testing for a progression-free survival HR of less than 1 would be at least 80%, while assuming exponentially distributed progression-free survival with median 4·35 months in the ramucirumab group and 3 months in the placebo group (ie, assuming a true underlying HR of 0·69). Thus the sample size of 450 also provided adequate power for the co-primary endpoint progression-free survival. The interim database lock was conducted at 272 progression-free survival events on Aug 5, 2019, and interim results were reviewed by an Independent Data Monitoring Committee (IDMC) to maintain sponsor blinding. Since the overall survival was immature at the interim analysis, in which only 59% (215/363) of overall survival events were observed, the success criterion (stratified overall survival HR <1 and progression-free survival nominal p<0·05) was not met at the interim analysis, and the study was continued until the final overall survival analysis, as per the IDMC's recommendation.

Efficacy analyses were done in the intention-to-treat (ITT) population, comprising all patients randomly assigned to study treatment. Safety analyses included all patients who received at least one dose of study treatment. Patient-reported outcomes were assessed in randomly assigned patients who provided both a baseline assessment and at least one post-baseline assessment.

For time-to-event endpoints including progression-free survival, overall survival, time to progression, and duration of objective response, the Kaplan-Meier product limit method was used to estimate the survival curves for each randomisation group and to generate median survival times, whose 95% CIs were computed by the Brookmeyer and Crowley method using log-log transformation. The stratified log-rank test was used to compare survival data between treatment groups. HRs and corresponding CIs were estimated in a stratified Cox proportional hazards model, with treatment group as the single covariate. Subgroup analyses for progression-free survival and overall survival based on baseline characteristics were conducted using the Cox proportional model. Objective response rate was reported with two-sided 95% CIs using the Clopper-Pearson method for each treatment group.

As a sensitivity analysis, a stepwise multivariate Cox regression model was fitted using predefined baseline characteristics to examine the effect of treatment on progression-free survival and overall survival after adjusting for significant prognostic factors. Overall survival was also analysed in the per-protocol population to explore the effect of protocol deviations. Additionally, sensitivity analyses for overall survival in which survival time was censored for users of different lines and types of post-discontinuation therapy were done to consider the

	Ramucirumab plus paclitaxel group (n=294)	Placebo plus paclitaxel group (n=146)
Age, years		
Median (range)	57·0 (22–84)	58·0 (18–79)
<65	225 (77%)	112 (77%)
≥65	69 (23%)	34 (23%)
Sex		
Male	205 (70%)	96 (66%)
Female	89 (30%)	50 (34%)
Country		
China	257 (87%)	135 (92%)
Malaysia	18 (6%)	7 (5%)
The Philippines	6 (2%)	2 (1%)
Thailand	13 (4%)	2 (1%)
ECOG performance status		
0	65 (22%)	31 (21%)
1	229 (78%)	115 (79%)
Site of primary tumour		
Gastric	259 (88%)	124 (85%)
Gastro-oesophageal junction	35 (12%)	22 (15%)
Tumour grade		
Well/moderately differentiated	49 (17%)	22 (15%)
Poorly differentiated/ undifferentiated	132 (45%)	73 (50%)
Unknown/missing	113 (38%)	51 (35%)
Primary tumour present		
Yes	178 (61%)	87 (60%)
No	116 (39%)	59 (40%)
Extent of disease		
Metastatic	286 (97%)	144 (99%)
Locally advanced	8 (3%)	2 (1%)

(Table 1 continues in next column)

(Continued from previous column)

	Ramucirumab plus paclitaxel group (n=294)	Placebo plus paclitaxel group (n=146)
Peritoneal metastasis		
Yes	103 (35%)	49 (34%)
No	191 (65%)	97 (66%)
Number of metastatic organs		
0–2	232 (79%)	112 (77%)
≥3	62 (21%)	34 (23%)
Weight loss over prior 3 months		
≥10%	25 (9%)	10 (7%)
<10%	267 (91%)	136 (93%)
Unknown	2 (<1%)	0 (0%)
Presence of ascites		
Yes	74 (25%)	38 (26%)
No	220 (75%)	108 (74%)
TTP on first-line therapy		
<6 months	165 (56%)	87 (60%)
≥6 months	129 (44%)	59 (40%)
Prior treatment with fluoropyrimidine–platinum		
Triplet: fluoropyrimidine–platinum with anthracycline	10 (3%)	7 (5%)
Doublet: fluoropyrimidine–platinum without anthracycline	284 (97%)	139 (95%)

Data are n (%) unless otherwise indicated. ECOG=Eastern Cooperative Oncology Group. TTP=time-to-progression.

Table 1: **Patient characteristics (intention-to-treat population)**

potential effect of treatment after discontinuation on overall survival.

EORTC QLQ-C30 and EQ-5D-3L data were scored according to developer guidelines and summarised by treatment group using descriptive statistics for each assessment time point.[13,14] The EQ-5D-3L index score was calculated using a weighting algorithm developed to represent Chinese population preferences for health states.[15]

SAS version 9.4 was used for statistical analysis. This trial is registered with ClinicalTrials.gov, number NCT02898077.

Role of the funding source

The study sponsor collaborated with investigators on protocol, study design, data collection, analysis, and interpretation, writing and preparation of this report.

Results

Between March 2, 2017, and June 30, 2020, 563 patients were screened; 440 eligible patients were randomly assigned to receive ramucirumab plus paclitaxel (n=294) or placebo plus paclitaxel (n=146) and were included in the efficacy analyses (figure 1). As of the data cut-off (June 30, 2020), 363 deaths had occurred and the median duration of follow-up was 8·05 months (IQR 4·67–12·29). Five (2%) patients in the ramucirumab plus paclitaxel group and three (2%) in the placebo plus paclitaxel group were still receiving study treatment at data cut-off.

Patient demographics and baseline characteristics were well balanced between the treatment groups (table 1). Of the 440 randomly assigned patients (median age 57·5 years [range 18–84]; 301 [68%] male), 392 (89%) were Chinese, 344 (78%) had an ECOG performance status of 1, and 152 (35%) had peritoneal metastasis. A large proportion of patients had poor prognostic factors, including poorly differentiated tumours, disease progression within 6 months after the start of first-line therapy, and metastatic disease, with 96 (22%) patients having at least three metastatic organs, 265 (60%) still having their primary tumour, 213 (48%) having liver metastases, and 112 (25%) having ascites.

After disease progression or the end of study treatment, the proportion of patients receiving systemic post-discontinuation therapy was similar between treatment groups (160 [54%] in the ramucirumab plus paclitaxel group and 82 [56%] in the placebo plus paclitaxel group). The most commonly used systemic post-discontinuation

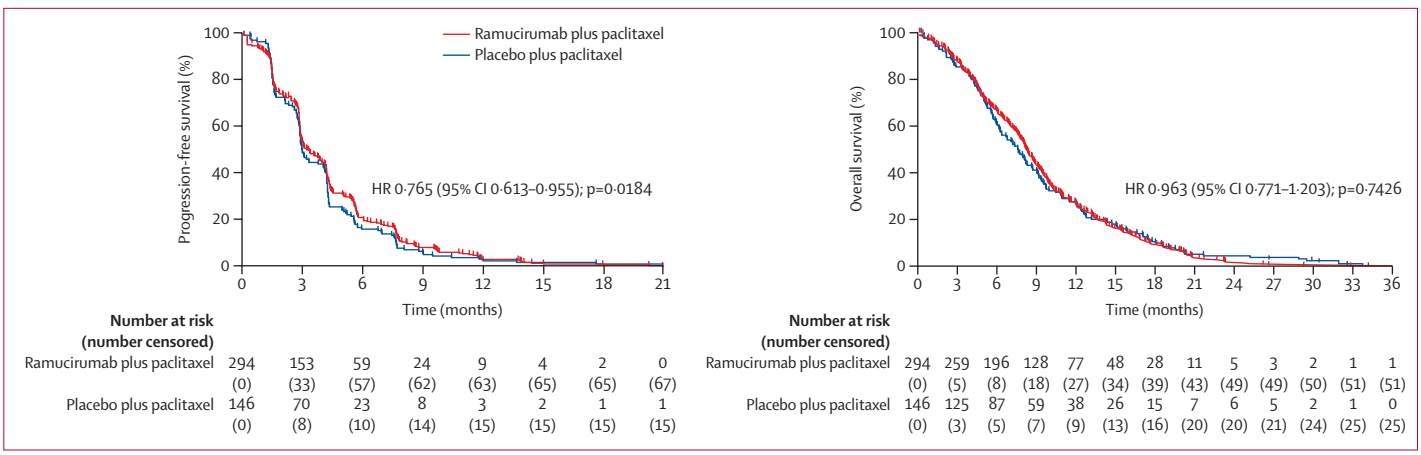

Figure 2: **Kaplan-Meier plots of progression-free survival and overall survival**
HR=hazard ratio.

	Ramucirumab plus paclitaxel group (n=294)	Placebo plus paclitaxel group (n=146)
Best overall response		
Complete response	0	2 (1·4%)
Partial response	78 (26·5%)	30 (20·5%)
Stable disease	148 (50·3%)	74 (50·7%)
Progressive disease	44 (15·0%)	31 (21·2%)
Not evaluable or not done	24 (8·2%)	9 (6·2%)
Objective response rate	26·5% (21·6–32·0)	21·9% (15·5–29·5)
Disease control rate	76·9% (71·6–81·6)	72·6% (64·6–79·7)

Data are n (%) or % (95% CI).

Table 2: **Best overall response (intention-to-treat population)**

therapies in both groups included chemotherapy, anti-angiogenesis therapy, and immunotherapy (appendix 2 p 3).

227 progression-free survival events occurred in the ramucirumab plus paclitaxel group and 130 in the placebo plus paclitaxel group. Progression-free survival was longer in the ramucirumab plus paclitaxel group compared with the placebo plus paclitaxel group (median 4·14 months [95% CI 3·71–4·30] *vs* 3·15 months [2·83–4·14]; stratified HR 0·765 [95% CI 0·613–0·955]; p=0·0184; figure 2). A sensitivity analysis of the progression-free survival data provided similar results after adjustment for prognostic factors (adjusted HR 0·714 [95% CI 0·572–0·892]; appendix 2 p 4). Progression-free survival was improved with ramucirumab plus paclitaxel in most subgroups (appendix 2 p 7).

There were 242 deaths in the ramucirumab plus paclitaxel group and 121 deaths in the placebo plus paclitaxel group. Median overall survival was 8·71 months (95% CI 7·98–9·49) in the ramucirumab plus paclitaxel group and 7·92 months (6·31–9·10) in the placebo plus paclitaxel group (HR 0·963 [95% CI 0·771–1·203], p=0·7426; figure 2). Sensitivity analyses of the overall survival data censoring users of post-discontinuation systemic therapy, adjusting for potential prognostic factors, and including an analysis of the per-protocol set, were consistent with the primary ITT analysis (appendix 2 p 5). Subgroup analyses of overall survival were generally consistent with the ITT population across nearly all subgroups (appendix 2 p 8).

Median time to progression was 4·3 months (95% CI 4·1–5·5) in the ramucirumab plus paclitaxel group and 4·1 months (2·9–4·2) in the placebo plus paclitaxel group. The proportion of patients who had an objective response or exhibited disease control was numerically higher in the ramucirumab plus paclitaxel group than in the placebo plus paclitaxel group (table 2). Median duration of response was longer in the ramucirumab plus paclitaxel group than the placebo plus paclitaxel group (4·34 months [95% CI 3·12–5·22] *vs* 2·83 months [2·56–4·14]).

Completion rates of the EORTC QLQ-C30 were high, with 405 (92%) patients (272 in the ramucirumab plus paclitaxel group and 133 in the placebo plus paclitaxel group) completing both baseline and post-baseline assessments. Baseline and short-term follow-up results for global quality of life from the QLQ-C30 and index scores from the EQ-5D-3L were similar between the treatment groups (appendix 2 p 6). Further details of QLQ-C30 results will be published elsewhere.

One patient in each of the randomised groups did not receive study medication and was excluded from the safety analyses; thus 438 patients (293 in the ramucirumab plus paclitaxel group and 145 in the placebo plus paclitaxel group) were included in the safety analyses. Median duration of treatment with ramucirumab was 14·0 weeks (IQR 8·0–24·0) in the ramucirumab plus paclitaxel group and 12·6 weeks (8·0–20·0) with placebo in the placebo plus paclitaxel group. Median relative dose intensity of ramucirumab was similar to that of placebo (98·4% [IQR 93·2–100·1] *vs* 99·6% [95·0–100·7]), and

	Ramucirumab plus paclitaxel group (n=293)				Placebo plus paclitaxel group (n=145)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
Any patient with a treatment-emergent adverse event	57 (19%)	120 (41%)	104 (35%)	9 (3%)	44 (30%)	63 (43%)	32 (22%)	4 (3%)
White blood cell count decreased	100 (34%)	106 (36%)	21 (7%)	0	65 (45%)	40 (28%)	2 (1%)	0
Neutrophil count decreased	64 (22%)	76 (26%)	83 (28%)	0	46 (32%)	33 (23%)	23 (16%)	0
Anaemia	150 (51%)	42 (14%)	3 (1%)	1 (<1%)	70 (48%)	22 (15%)	2 (1%)	0
AST increased	92 (31%)	6 (2%)	2 (<1%)	0	32 (22%)	3 (2%)	0	0
Proteinuria	92 (31%)	6 (2%)	0	0	30 (21%)	1 (<1%)	0	0
Hypoalbuminaemia	88 (30%)	1 (<1%)	0	0	30 (21%)	1 (<1%)	0	0
ALT increased	79 (27%)	9 (3%)	0	0	30 (21%)	1 (<1%)	0	0
Platelet count decreased	77 (26%)	7 (2%)	4 (1%)	0	16 (11%)	1 (<1%)	0	0
Alopecia	83 (28%)	0	0	0	55 (38%)	0	0	0
Decreased appetite	75 (26%)	6 (2%)	0	0	31 (21%)	2 (1%)	0	0
Malaise	73 (25%)	1 (<1%)	0	0	37 (26%)	0	0	0
Diarrhoea	65 (22%)	4 (1%)	0	0	20 (14%)	1 (<1%)	0	0
Blood bilirubin increased	56 (19%)	7 (2%)	2 (<1%)	0	30 (21%)	5 (3%)	0	0
Hypertension	44 (15%)	21 (7%)	0	0	16 (11%)	9 (6%)	0	0
Pyrexia	55 (19%)	1 (<1%)	0	0	24 (17%)	0	0	0
Weight decreased	58 (20%)	0	0	0	21 (14%)	1 (<1%)	0	0
Nausea	53 (18%)	0	0	0	22 (15%)	0	0	0
Epistaxis	53 (18%)	0	0	0	9 (6%)	0	0	0
Vomiting	46 (16%)	6 (2%)	0	0	21 (14%)	3 (2%)	1 (<1%)	0
Constipation	50 (17%)	0	0	0	26 (18%)	0	0	0
Abdominal distension	44 (15%)	3 (1%)	0	0	15 (10%)	0	0	0
Hypocalcaemia	45 (15%)	0	0	0	14 (10%)	0	0	0
Abdominal pain	35 (12%)	5 (2%)	0	0	23 (16%)	3 (2%)	0	0
Insomnia	35 (12%)	2 (<1%)	0	0	10 (7%)	1 (<1%)	0	0
Hypoesthesia	35 (12%)	1 (<1%)	0	0	22 (15%)	0	0	0
GGT increased	29 (10%)	6 (2%)	1 (<1%)	0	10 (7%)	3 (2%)	1 (<1%)	0
Oedema peripheral	35 (12%)	0	0	0	9 (6%)	0	0	0
Blood alkaline phosphatase increased	25 (9%)	9 (3%)	0	0	14 (10%)	3 (2%)	0	0
Hyponatraemia	21 (7%)	11 (4%)	0	0	16 (11%)	5 (3%)	2 (1%)	0
Hypokalaemia	24 (8%)	7 (2%)	1 (<1%)	0	10 (7%)	5 (3%)	1 (<1%)	0
Back pain	32 (11%)	0	0	0	14 (10%)	0	0	0
Cough	30 (10%)	0	0	0	9 (6%)	0	0	0

Data are n (%). ALT=alanine aminotransferase. AST=aspartate aminotransferase. GGT=gamma glutamyl transferase.

Table 3: **Treatment-emergent adverse events occurring in at least 10% of patients in the ramucirumab plus paclitaxel group, irrespective of causality**

the median relative dose intensity of paclitaxel was slightly lower in the ramucirumab plus paclitaxel group than in the placebo plus paclitaxel group (80·4% [IQR 70·1–91·1] *vs* 88·9% [80·0–95·4]). Dose reductions of ramucirumab occurred in 17 (6%) of 293 patients in the ramucirumab plus paclitaxel group, and of placebo in six (4%) of 145 patients in the placebo plus paclitaxel group. Paclitaxel dose reductions occurred in 77 (26%) patients in the ramucirumab plus paclitaxel group and in 18 (12%) patients in the placebo plus paclitaxel group. Disease progression was the most common reason for treatment discontinuation in both treatment groups (185 [63%] patients in the ramucirumab plus paclitaxel group *vs* 108 [74%] in the placebo plus paclitaxel group); 21 (7%) and seven (5%) patients, respectively, discontinued study treatment due to adverse events.

The incidence of treatment-emergent adverse events was similar between groups (290 [99%] patients in the ramucirumab plus paclitaxel group and 143 [99%] in the placebo plus paclitaxel group). Grade 3 or 4 treatment-emergent adverse events were observed in 224 (76%) patients in the ramucirumab plus paclitaxel group and 95 (66%) in the placebo plus paclitaxel group. The most commonly reported grade 3 or 4 treatment-emergent adverse events were decreased neutrophil count, decreased white blood cell count, and anaemia (table 3).

	Ramucirumab plus paclitaxel group (n=293)				Placebo plus paclitaxel group (n=145)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
Liver injury	163 (56%)	13 (4%)	7 (2%)	0	70 (48%)	10 (7%)	1 (<1%)	0
Bleeding/ haemorrhage events	91 (31%)	7(2%)	2 (<1%)	0	23 (16%)	5 (3%)	0	0
Gastrointestinal haemorrhage events	24 (8%)	6 (2%)	2 (<1%)	0	9 (6%)	4 (3%)	0	0
Proteinuria	93 (32%)	6 (2%)	0	0	30 (21%)	1 (<1%)	0	0
Hypertension	47 (16%)	24 (8%)	0	0	17 (12%)	9 (6%)	0	0
Infusion-related reaction*	14 (5%)	2 (<1%)	0	0	2 (1%)	0	0	0
Venous thromboembolic events	6 (2%)	0	1 (<1%)	0	4 (3%)	0	1 (<1%)	0
Congestive heart failure	3 (1%)	2 (<1%)	0	0	1 (<1%)	0	0	0
Arterial thromboembolic events	2 (<1%)	0	0	1 (<1%)	1 (<1%)	1 (<1%)	0	0
Fistula	2 (<1%)	0	1 (<1%)	0	0	0	0	0
Gastrointestinal perforation	1 (<1%)	0	0	0	0	0	0	0
Healing complication	0	0	0	0	0	0	0	0
Posterior reversible encephalopathy syndrome	0	0	0	0	0	0	0	0

Data are n (%). *Potential immediate hypersensitivity reactions (narrow events) which represent infusion-related reactions.

Table 4: **Treatment-emergent adverse events of special interest**

Febrile neutropenia was reported by 18 (6%) patients in the ramucirumab plus paclitaxel group and one (<1%) patient in the placebo plus paclitaxel group. Findings from the analysis of laboratory data did not reveal any new or increased safety concerns (data not shown).

Serious adverse events were reported in 99 (34%) patients in the ramucirumab plus paclitaxel group and 37 (26%) in the placebo plus paclitaxel group. Treatment-emergent adverse events leading to death occurred in nine (3%) patients in the ramucirumab plus paclitaxel group and four (3%) in the placebo plus paclitaxel group. Four (1%) patients in the ramucirumab plus paclitaxel group had adverse events leading to death that were deemed to be related to study treatment, which were multiple organ dysfunction syndrome, respiratory failure, anaemia, and cerebral infarction. Five deaths in this group were deemed unrelated to study treatment (pleural effusion, sudden cardiac death, death by suicide, sepsis, and neutropenic sepsis). Two (1%) patients in the placebo plus paclitaxel group had adverse events leading to death attributed to study treatment, which were sudden death and respiratory failure. Two deaths in this group (craniocerebral injury and interstitial lung disease) were deemed unrelated to study treatment.

Adverse events of special interest are summarised in table 4. Incidences of grade 3 or worse adverse events of special interest were similar between the two groups, with no grade 3 or worse gastrointestinal perforation, healing complication, or posterior reversible encephalopathy syndrome. Incidences of grade 4 and 5 adverse events of special interest were low in both groups, with similar incidences of liver injury, haemorrhage, venous thromboembolic events, arterial thromboembolic events, and fistula.

Discussion

RAINBOW-Asia, as a bridging study of RAINBOW, met its predefined success criterion (stratified HR for overall survival <1 and nominal p-value for progression-free survival <0·05) in accordance with requirements for the bridging study from the Chinese regulatory authority. Efficacy data demonstrated consistency with the results of the RAINBOW study.[8] Progression-free survival was numerically longer with ramucirumab plus paclitaxel than with placebo plus paclitaxel in Asian, predominantly Chinese, patients with advanced gastric or GEJ adenocarcinoma that had progressed after first-line chemotherapy; overall survival was also numerically longer with ramucirumab and paclitaxel, although there was no significant difference between groups. The treatment effect was consistent across most of the patient subgroups analysed.

Although the design of the RAINBOW-Asia study was similar to that of RAINBOW, it should be noted that differences exist between the two studies. Specifically, RAINBOW-Asia was a bridging study designed to show consistency in efficacy with RAINBOW, rather than to show statistical superiority of treatment with ramucirumab plus paclitaxel over placebo plus paclitaxel. RAINBOW-Asia had co-primary endpoints of progression-free survival and overall survival, and randomisation was stratified by ECOG performance status and presence of peritoneal metastases, as both factors were found to be independent prognostic factors for overall survival in previous studies.[16,17] Furthermore, one of the requirements for eligibility for this study was that patients had to have at least one measurable lesion.

The patients in this study were generally reflective of the overall population of Chinese patients with advanced gastric or GEJ adenocarcinoma enrolled in other clinical trials.[18–20] There were different prognosis patterns at baseline in RAINBOW and RAINBOW-Asia. Overall, the proportion of patients with GEJ as the primary tumour site was lower in RAINBOW-Asia than in RAINBOW; furthermore, the proportion of patients with time to progression of less than 6 months on first-line therapy was lower, the proportion of patients with zero to two metastatic sites was higher, and the incidences of peritoneal metastases and ascites were lower in RAINBOW-Asia than

in RAINBOW.[8] Patients in RAINBOW-Asia also had worse performance status at baseline. The differences in baseline prognostic patterns between these patient populations has been noted previously.[21,22]

In this study, the observed HR for progression-free survival was 0·765 (95% CI 0·613–0·955). Median progression-free survival was 4·14 months in the ramucirumab plus paclitaxel group versus 3·15 months in the placebo plus paclitaxel group. The improvement in progression-free survival with ramucirumab plus paclitaxel treatment was observed across most subgroups (eg, time to progression on first-line therapy, liver metastasis, use of post-discontinuation systemic therapy). The findings of progression-free survival improvement in the present study were consistent with those in the global RAINBOW study.[8]

In the present study, the observed stratified HR for overall survival was 0·963 (95% CI 0·771–1·203). As a bridging study, the HR for overall survival point estimate met the prespecified requirement of being less than 1 to be deemed consistent with the overall findings of the RAINBOW trial by the regulatory authorities in China, when interpreted with progression-free survival results. However, statistical superiority of ramucirumab plus paclitaxel was not demonstrated. Global trials have shown a statistically significant and clinically meaningful improvement in median overall survival with ramucirumab in patients with advanced gastric or GEJ adenocarcinoma in the second-line setting, including RAINBOW (9·6 months *vs* 7·4 months).[8,11] A possible explanatory factor for the difference in overall survival results found between RAINBOW and RAINBOW-Asia is the greater use of further lines of post-discontinuation systemic therapy in our study population. Extensive use of post-discontinuation systemic therapy in Asian and Japanese patients with gastric cancer has been reported previously.[21–23] Subgroup analyses of east Asians versus non-east Asians[21] and Japanese versus western patients[23] in RAINBOW showed that while use of post-discontinuation systemic therapy was similar regardless of study treatment, it was more common among east Asian (67%) and Japanese patients (75%) than in non-east Asian (37%) and western patients (37%).[21,23] In the present study, a similar proportion of patients in the ramucirumab plus paclitaxel group compared with the placebo plus paclitaxel group received at least one subsequent line of post-discontinuation systemic therapy (54% *vs* 56%), including anti-angiogenesis agents (used by around a quarter of patients). Exploratory survival analyses evaluating the effect of use of post-discontinuation systemic therapy on survival showed that the median overall survival was longer in patients who received post-discontinuation systemic therapy versus those who did not, regardless of the study treatment group. A higher percentage of east Asian patients in the placebo plus paclitaxel group of RAINBOW received post-discontinuation systemic therapy compared with non-east Asian patients in RAINBOW and patients in RAINBOW-Asia (66% *vs* 36% *vs* 55%); consistent with this observation, median overall survival in the placebo plus paclitaxel group was 10·5 months among east Asian patients in RAINBOW, compared with 5·9 months in non-east Asian patients in RAINBOW and 7·92 months in the current study.[21] The high uptake of further lines of treatment and relatively long survival could have diluted any potential survival benefit associated with ramucirumab.[24,25] A sensitivity analysis of overall survival, censoring users of post-discontinuation systemic therapy by the number of lines of post-discontinuation systemic therapy, showed increasing HRs (albeit not significant) with increasing lines of post-discontinuation systemic therapy, which also suggests that the intensity of post-discontinuation systemic therapy might have confounded the magnitude of the overall survival benefit.

Overall, the safety profile of ramucirumab plus paclitaxel in RAINBOW-Asia was consistent with that observed in RAINBOW,[8,21,23] and generally in line with the known safety profiles of the individual treatment components,[26] with no unexpected safety events reported. Decreased neutrophil count was the most common grade 3 or worse treatment-emergent adverse event in both groups, although it occurred at a higher incidence in the ramucirumab plus paclitaxel group than in the placebo plus paclitaxel group. Nonetheless, the incidence of febrile neutropenia was low in both treatment groups. The mechanism responsible for the decrease in neutrophil count seen with ramucirumab is unknown. However, these incidences were comparable with those reported in previous studies of paclitaxel.[27–30]

The main limitation of the present study is the lack of biomarker research. Further, we did not collect data for the histological subtype (Lauren classification) of gastric or GEJ adenocarcinoma at baseline.

In conclusion, the results of RAINBOW-Asia did not demonstrate statistical superiority of ramucirumab plus paclitaxel over placebo plus paclitaxel, but did show efficacy of ramucirumab plus paclitaxel as second-line therapy in a predominantly Chinese population with advanced gastric or GEJ adenocarcinoma based on predefined criterion. Ramucirumab's safety profile was consistent with that observed in previous studies, and no new safety concerns were raised. The efficacy data and the safety profile in RAINBOW-Asia and RAINBOW demonstrate a favourable risk–benefit profile and provide evidence suggesting that ramucirumab plus paclitaxel might be an effective treatment option for Chinese patients with advanced gastric or GEJ adenocarcinoma in the second-line setting.

Contributors

R-HX and LS made substantial contributions to the conception and design of the work. R-HX, YQZ, HMP, JFF, TZ, TSL, YRQ, SKQ, XLY, BRL, YB, NY, PJV, ST, and LS contributed to data acquisition. WLZ contributed to data analysis. CZ and WLZ contributed to data interpretation. All authors reviewed or revised the manuscript and approved the final version. All authors had full access to all the data in the study, vouch for the integrity of the data, and had final responsibility for the decision to submit the manuscript for publication.

Declaration of interests
CZ and WLZ are employees of Eli Lilly and Company. LS has received grants from Beijing Xiantong Biomedical Technology, Qilu Pharmaceutical, ZaiLab Pharmaceutical (Shanghai), Beihai Kangcheng (Beijing) Medical Technology, and Jacobio Pharmaceuticals; consulting fees from Merck Sharp & Dohme, Merck, Mingji Biopharmaceutical, Haichuang Pharmaceutical, Herbour Biomed, and Boehringer Ingelheim; payment for speakers bureaus from Hutchison Whampoa, Hengrui, ZaiLab Pharmaceutical (Shanghai), and CSTONE Pharmaceutical; and has participated as an advisory board member for Rongchang Pharmaceutical, ZaiLab Pharmaceutical (Shanghai), CSTONE Pharmaceutical, and Bristol Myers Squibb. All other authors declared no competing interests.

Data sharing
Lilly provides access to all individual participant data collected during the trial, after anonymisation, except for pharmacokinetic or genetic data. Data are available for request 6 months after the indication studied have been approved in the US and EU and after primary publication acceptance, whichever is later. No expiration date of data requests is currently set once data are made available. Access is provided after a proposal has been approved by an independent review committee identified for this purpose and after receipt of a signed data sharing agreement. Data and documents, including the study protocol, statistical analysis plan, clinical study report, blank or annotated case report forms, will be provided in a secure data sharing environment. For details on submitting a request, see the instructions provided at www.vivli.org.

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